

SLEEVEDRIFT · DESIGN DECISIONS REGISTER

SleeveDrift design decisions register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/sleevedrift		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-04	Amish Chadha	Open decision 1 (R3 target) decided by Amish (round-3 decision 4A, SVD-DDR-004) and moved to Decisions made; no open decisions
0.2	2026-10-03	Amish Chadha	Amish decided the three open items as recommended (24B, 25A, 26C; SVD-DDR-003); one new question on restating R3 proposed
0.1	2026-10-03	Amish Chadha	Register opened; design decisions made under Amish's pre-approvals of 2026-10-03; three requirement decisions proposed for Amish

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Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` or in `docs/REVIEW.md`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list decisions.

Safety: Several decisions below set SleeveDrift's safety case for people working inside a rescue pipe (overload limit, gas monitoring, air, ring fixing, casualty evacuation). Each took the conservative option; the evidence that would relax it is in SVD-DDR-001, Table 1.

■ Open decisions

None. The three items of 2026-10-03 and the R3 restatement (decided 2026-10-04) are all under Decisions made.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	Pipe bore, wall and internal weld beads of the pipe used for trials	The ring is 760 mm OD with 20 mm clearance; a bead or a smaller bore changes the fit	SVD-CAL-001, A; R1
2	Winch power ratios, base bolt pattern and line rating	Handle forces assume ratios of about 13 and 40; the trestle top plates are drilled to the base	SVD-CAL-001, C; BOM line 17
3	Release load of the breakaway swivel pins bought	Sets the 2.5 kN limit that every part is sized on	SVD-CAL-001, C; R11
4	Sheave bore, bush and rating	The pin is 25 mm; the sheave must be rated at least 1,000 kg	BOM line 6
5	Friction of the jacking pads on the actual pipe surface	The ring's hold against the sheave pull assumes 0.20	SVD-CAL-001, C8
6	Rope breaking strength and splice efficiency	The factor of 6.5 assumes 18 kN and 90 % at the splice	SVD-CAL-001, C6
9	Drill torque, clutch range and no-load speed in high gear; winch bit to fit the socket	The drive assumes 120 rpm at the socket and a clutch that can be set to about 35 N m	SVD-CAL-001, D4 and D6; BOM lines 30 and 31
7	Blower curve at 525 Pa	The air figures assume 0.27 m ³ /s at the blower	SVD-CAL-001, F; R5
8	Holding force of the magnet hangers on painted or rusty pipe	They carry the duct and the cable	BOM line 22

■ Value engineering

Value-engineering target: USD 5,000 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 10,804.00 (USD 5,804.00 over the target), including the drill drive (USD 700) and the spare rope set (USD 236.50) decided on 2026-10-03. Main cost drivers and savings worth trying:

- The two self-tailing winches (USD 2,300) are the largest line; reconditioned winches of the same size are often about half the new price.
- The four-gas monitors (USD 1,100), the blower and duct (USD 1,515) and the telephones (USD 860) are kit that rescue services usually already hold; borrowing them for trials removes about USD 3,500 from the prototype cost without changing the design.
- The made steel (ring, trestles, sledges) is under USD 1,500; plate rolling is the main cost there.
- The drill kits (USD 500) can be bought in the battery platform a rescue service already uses, so its batteries and chargers are shared.
- Savings not worth taking: the breakaway swivels, rope and slings carry the safety case and stay at rated grades.

■ Decisions made

Pre-approval quotes: Q1, Amish 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost." Q2, Amish 2026-10-03: "Proceed with the remaining 15 scaffolds" (under the same pre-approval). Q3, Amish 2026-10-03: "i agree with all the 46 recommendations you provided. please proceed."

Date	Decision	Decided by	Record
2026-10-03	Design-case pipe 800 x 10 mm, 780 mm bore	Amish, Q1 and Q2	SVD-DDR-001, item 1
2026-10-03	Ring held by jacking screws with swivel pads; nothing drilled or welded to the pipe	Amish, Q1 and Q2	SVD-DDR-001, item 2
2026-10-03	Shuttle train between a pull rope and a tail rope, in place of a circulating loop	Amish, Q1 and Q2	SVD-DDR-001, item 3
2026-10-03	Two two-speed self-tailing winches at the portal; SiltHaul capstan not used	Amish, Q1 and Q2	SVD-DDR-001, items 4, 12
2026-10-03	Breakaway swivels at 2.5 kN at both ends of the train; requirement R11 added	Amish, Q1 and Q2	SVD-DDR-001, item 5
2026-10-03	Two four-gas monitors in the kit; requirement R12 added	Amish, Q1 and Q2	SVD-DDR-001, item 6
2026-10-03	Forcing ventilation through 200 mm layflat, about 15 m ³ /min at the face	Amish, Q1 and Q2	SVD-DDR-001, item 7
2026-10-03	Sound-powered telephones with a pull-cord bell	Amish, Q1 and Q2	SVD-DDR-001, item 8
2026-10-03	Bought roll-up stretcher with a made spreader bar; casualty hauled hand over hand through winch A	Amish, Q1 and Q2	SVD-DDR-001, items 9, 10
2026-10-03	First co-design candidate to approach: India's National Disaster Response Force with the Jhansi hand miners (not agreed)	Amish, Q1 and Q2	SVD-DDR-001, item 11
2026-10-03	CalRig as the first candidate rig for the hood test and proof loads; HatchSide as the first reference for air and gas rules	Amish, Q1 and Q2	SVD-DDR-001, item 12
2026-10-03	budget_usd kept at 5,000 as a value-engineering target	Amish, Q1 ("I also accept any cost overruns or variations from the assumed scope cost.")	SVD-DDR-001, item 13
2026-10-03	Design for construction: the twelve changes of SVD-DDR-002	Amish, Q1 and Q2	SVD-DDR-002
2026-10-03	Appearance model additions for renders: pipe window and platform cropped, a 1.75 m mannequin beside the pipe	Amish, Q1 and Q2	docs/REVIEW.md, TRL 3
2026-10-03	R3: a right-angle electric drill drive on each portal winch (about 0.42 m ³ /h, about USD 700); nothing extra inside the pipe; drill safety kept: guarding, dead-man trigger, breakaway swivels (item 24, option B)	Amish, Q3	SVD-DDR-003
2026-10-03	R9: inspect both ropes each shift and carry a spare set (item 25, option A)	Amish, Q3	SVD-DDR-003
2026-10-03		Amish, Q3	SVD-DDR-003

Date	Decision	Decided by	Record
	R1: the 600 mm variant waits for the co-design partner (item 26, option C)		
2026-10-04	R3 (4A): restated as "moves loose spoil faster than the face produces it; at least 0.4 m ³ /h" (hand-dug faces about 0.27 m ³ /h); met at 0.42 m ³ /h	Amish: "For round 3, I agree with all your proposed recommendations"	SVD-DDR-004